

Pyuria prediction performance brief

Leakage-safe pediatric EHR model: LightGBM + multiscale LSM stack

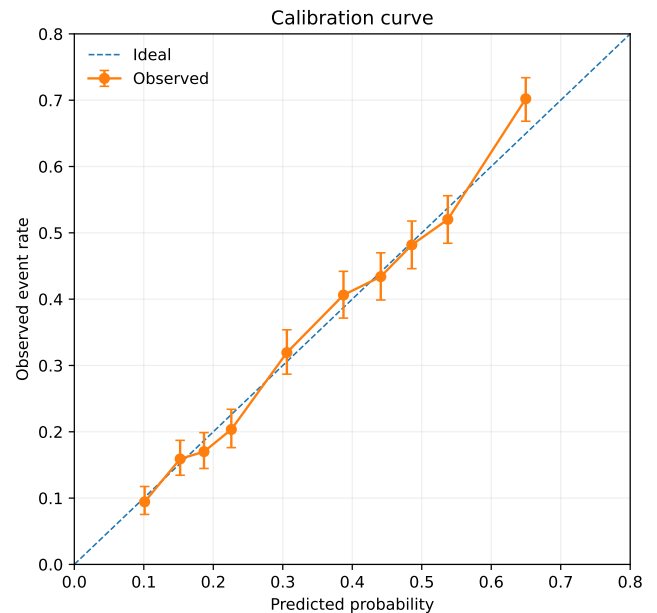
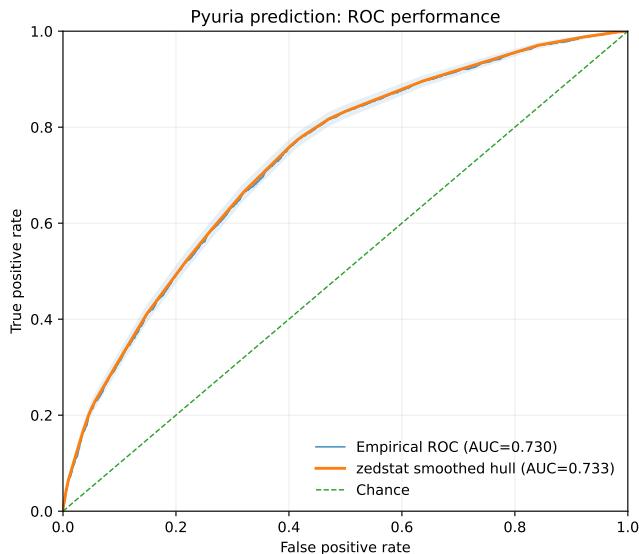
Cohort and design. Each row is a urinalysis event; the target is microscopic pyuria (> 5 WBC/HPF). The fixed patient-level split uses seed 20261006, with 11,297 training events and 7,417 held-out test events. Test prevalence is 34.9%. The final model uses only information judged reliably available before the UA index time.

Final models. LightGBM uses 110 leakage-safe EHR predictors. A 60-feature categorical representation is used for four Large Science Models (LSMs), trained with $\alpha = 0.1$ on persistence-peeled residual populations of 11,297, 3,521, 1,124, and 376 rows. The final logistic stack combines LightGBM probability with the four masked-target LSM conditionals.

Model	Held-out AUC
Multiscale LSMs only	0.7023
LightGBM	0.7283
LightGBM + root LSM	0.7298
LightGBM + LSM depths 0–1	0.7300
LightGBM + LSM depths 0–3	0.7304

zedstat evaluation. Empirical held-out AUC is 0.7304 (95% CI 0.7184–0.7420). The zedstat smoothed upper-hull AUC is **0.7329** (95% CI 0.7204–0.7454). At the smoothed Youden operating point (threshold ≈ 0.3175):

Metric	Estimate	95% CI
Sensitivity	0.773	0.756–0.788
Specificity	0.586	0.572–0.600
PPV	0.500	0.485–0.516
NPV	0.828	0.814–0.841
Accuracy	0.651	0.636–0.666
LR+	1.866	1.767–1.970
LR-	0.388	0.353–0.426



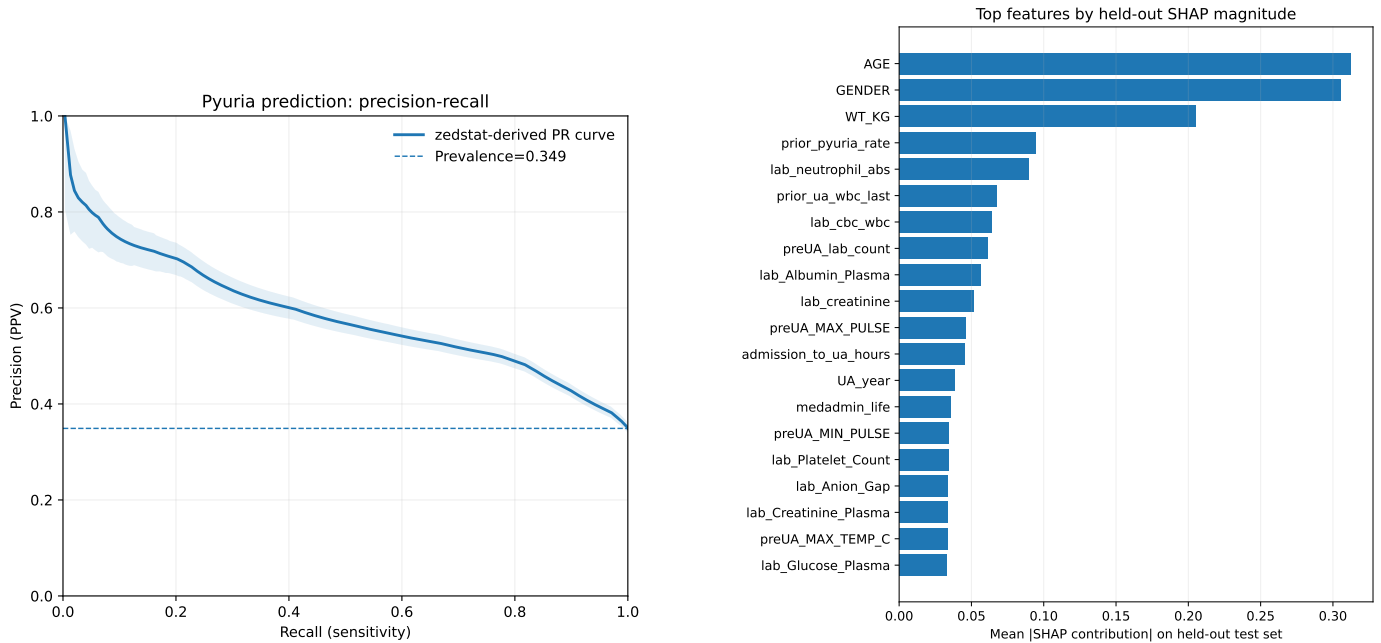
Calibration. Brier score is 0.1934 (95% CI 0.1894–0.1974). Calibration intercept is 0.0426 (95% CI -0.0197–0.1053) and slope is 1.064 (95% CI 0.999–1.134), both compatible with the ideal values 0 and 1. The highest risk decile shows mild underprediction (mean predicted risk $\approx$ 0.650 vs observed event rate $\approx$ 0.702).

High-specificity operating points.

Specificity	Sensitivity	PPV	LR+
0.90	0.316	0.629	3.16
0.95	0.213	0.696	4.27
0.99	0.0668	0.782	6.68

Most important predictors. Held-out SHAP magnitude and LightGBM gain both identify age, gender, weight, prior pyuria history, prior urine-WBC state, CBC/inflammatory measurements, pre-UA laboratory burden, albumin, creatinine, pulse, and admission-to-UA timing as dominant contributors. The top held-out SHAP variables are:

Rank	Feature	Mean SHAP
1	AGE	0.312
2	GENDER	0.306
3	WT_KG	0.205
4	prior_pyuria_rate	0.0947
5	lab_neutrophil_abs	0.0894
6	prior_ua_wbc_last	0.0677
7	lab_cbc_wbc	0.0636
8	preUA_lab_count	0.0614
9	lab_Albumin_Plasma	0.0564
10	lab_creatinine	0.0512



Interpretation. Richer, carefully censored pre-UA EHR data produced the largest performance gain. Multiscale LSM conditionals add a smaller but consistent increment beyond LightGBM. The analysis is retrospective and should be externally validated before clinical use.